A Prediction of Darkness

During the early 1930s, Fritz Zwicky, a professor of astronomy at the California Institute of Technology (a famously caustic scientist whose appreciation for symmetry led him to call his colleagues spherical bastards because, he explained, they were bastards any way you looked at them[19]), realized that the outlying galaxies in the Coma cluster, a collection of thousands of galaxies some 370 million light-years from earth, were moving too quickly for their visible matter to muster an adequate gravitational force to keep them tethered to the group. Instead, his analysis showed that many of the fastest-moving galaxies should be flung clear of the cluster, like water droplets thrown off a spinning bicycle tire. And yet none were. Zwicky conjectured that there might be additional matter permeating the cluster that did not give off light but supplied the additional gravitational pull necessary to hold the cluster together. His calculations showed that if this explanation was right, the vast majority of the cluster's mass would comprise this nonluminous material. By 1936, corroborating evidence was found by Sinclair Smith of the Mount Wilson observatory, who was studying the Virgo cluster and came to a similar conclusion. But since both men's observations, as well as a number of subsequent others, had various uncertainties, many remained unconvinced that there was voluminous unseen matter whose gravitational pull was keeping the groups of galaxies together.

Over the next thirty years observational evidence for nonluminous matter continued to mount,[20] but it was the work of the astronomer Vera Rubin from the Carnegie Institution of Washington, together with Kent Ford and others, that really clinched the case. Rubin and her collaborators studied the movements of stars within numerous spinning galaxies and concluded that if what you see is what there is, then many of the galaxy's stars should be routinely flung outward. Their observations showed conclusively that the visible galactic matter could not exert a gravitational grip anywhere near strong enough to keep the fastest-moving stars from breaking free. However, their detailed analyses also showed that the stars *would* remain gravitationally tethered if the galaxies they inhabited were immersed in a giant ball of nonluminous matter (as in Figure 10.5), whose total mass far exceeded that of the galaxy's luminous material. And so, like an audience that infers the